Presented by: Group 8

HOSPITAL MANAGEMENT SYSTEM

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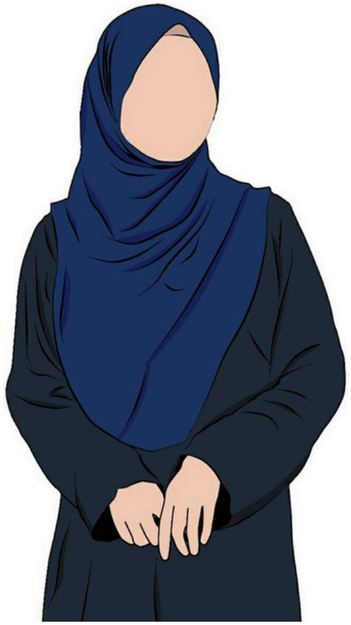

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INTRODUCTION

The Hospital Management System (HMS) database is designed to make hospital work easier by organizing patient records, appointments, bills, and staff details in one place.

This system will help reduce mistakes, save time, and give quick, secure access to important information, improving patient care and daily operations.

END USERS

Doctors

Access patient records, track history, and update treatments.

Medical Staff

Manage patient care and monitor vital information.

Admin Staff

Handle appointments, registrations, and billing.

Hospital Management

Oversee operations and generate reports.

Pharmacists

Track Inventory and merge prescription

Patients

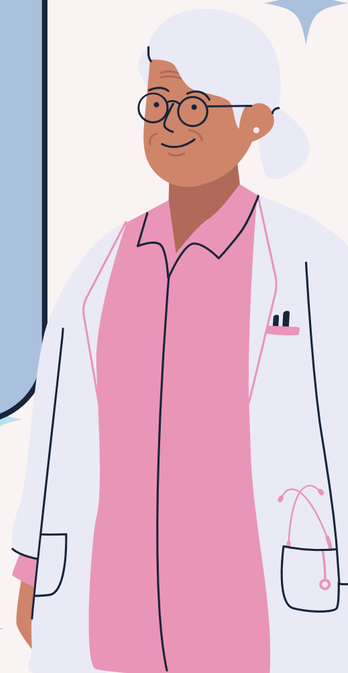
Book appointments, access medical records, and pay bills.

DATA SOURCE

Patient Data: Personal info, medical history, treatments, and prescriptions.

Clinical Data: Lab results, imaging reports, and vital signs.

Operational Data: Appointments, bed availability, and staff schedules.

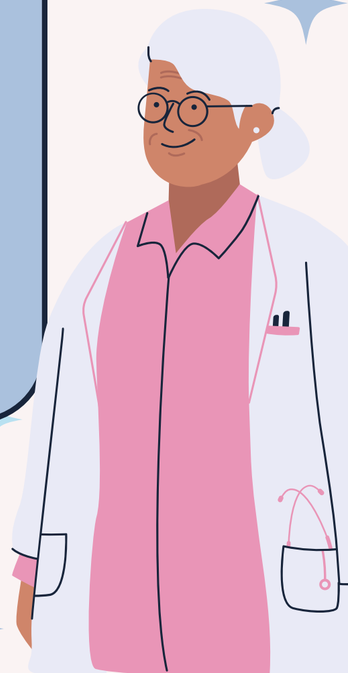


DATA SOURCE

Financial Data: Billing, insurance claims, and payments.

Inventory Data: Medical supplies, equipment, and pharmaceuticals.

External Data: Information from insurance providers and health databases.



Overall Working of the Hospital Management System

1. Patient Registration:
Assigns a unique ID to each patient.
Records personal and medical details.

2. Appointment Scheduling:
Manages doctors' schedules and patient appointments.
Sends reminders via SMS or email.

3. Clinical Workflow:
Doctors access records, diagnose, and prescribe treatments.
Nurses update vitals; lab staff upload test results.

4. Billing and Insurance:
Generates bills for treatments and services.
Processes and tracks insurance claims.

5. Inventory Management:
Monitors medical supplies and equipment.
Generates alerts for low stock or expired items.

6. Reporting and Analytics:
Creates performance, patient, and financial reports.
Supports data-driven decision-making.

7. Data Security and Compliance:

Encrypts patient data for secure storage.Ensures compliance with HIPAA, GDPR, and other regulations.

8. Integration with External Systems

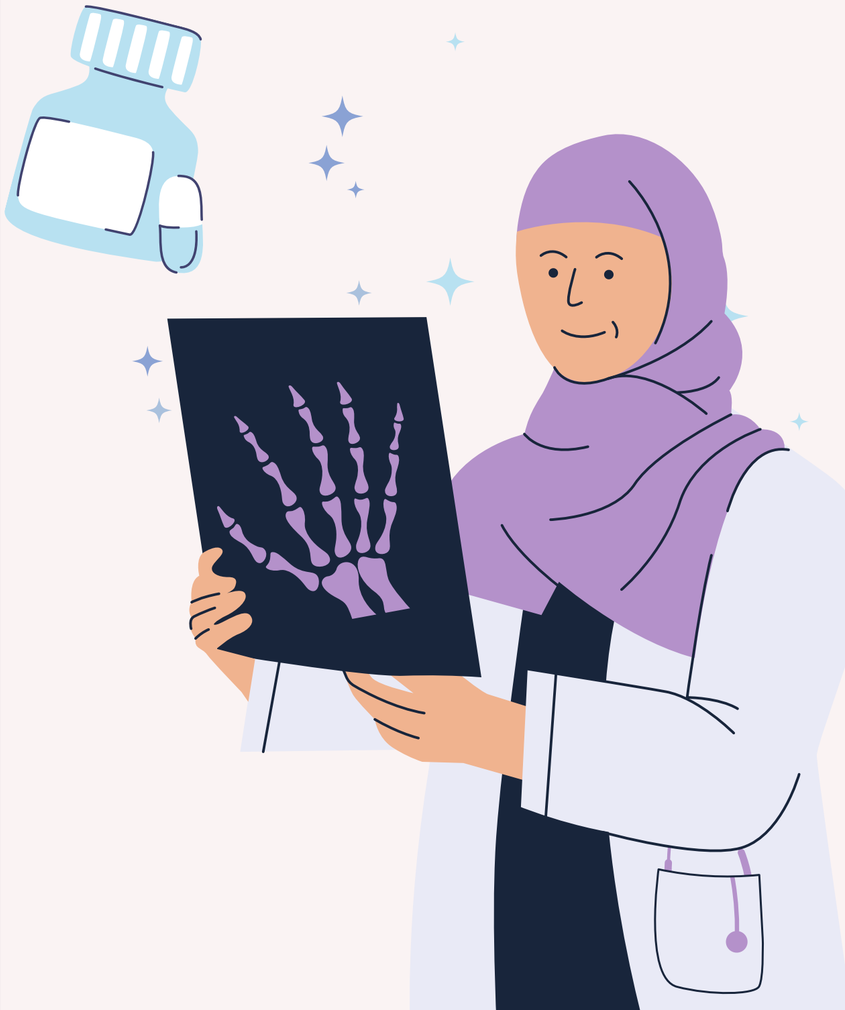
Connects with insurance providers, health portals, and diagnostic labs.Facilitates seamless data exchange.

9. Role Management:

Assigns access levels based on roles Ensures sensitive data is accessible only to authorized personnel.

Key Features of HMS:

- **Centralized database** for hospital operations.
- **Real-time** access to patient records.
- **Automated workflows** to reduce errors.
- **Enhanced communication** between departments.
- **Improved patient care** and satisfaction.
- **Increased efficiency** and cost reduction.



Thank you for your attention